

STATE OF THE ART · THESIS MORO

Guided Robotic Systems for Ultrasound & Breast Biopsy

Systematic analysis of the robotics literature underpinning the thesis: ultrasound lesion tracking (impedance control, visual servoing, planning) and ultrasound-guided breast biopsy (complete systems, force/position control, deformation compensation).

PAPERS

15 read in full

TASKS

Tracking · Biopsy

EXTRACTION CRITERIA

Study · Robot · Control · Benchmark

SOURCE

Scopus + IEEE Xplore

Università di Trento · DISI · Supervisors: E. Lamon (robotics) · S. Tomassini (medical imaging) · Extraction format per Lamon brief 17/12/2025

01 Ultrasound Lesion Tracking

CONTINUOUS · IMAGING-GUIDED · DEFORMABLE CONTACT

1a Impedance Control + probe–tissue contact

SCOPUS ("robot*") AND ("ultrasound") AND ("impedance control") AND ("tracking")

#	Authors, Year	Study	Robot	Control method	Key result	Benchmark	DOI
1	Ipsen et al. 2021 <i>Phys. Med. Biol.</i>	In vivo 5 volunteers, 30 min	KUKA LBR iiwa 7 R800 7-DOF	Asymmetric Cartesian Impedance: $k_z=0$ N/m (compliant along probe, passive breathing), $k_{x,y}=100$ N/m (stiff laterally). Force target 10 N, safety 25 N.	Median force 10.0±0.1 N over 10 sessions. Entropy 0.96±0.01 stable 30 min. Liver excursion up to 45 mm passively compensated. Breast ref: 3–5 N.	Yes — 10 sessions, 2 organs	10.1088/1361-6560/abf277
2	Fu et al. 2024 <i>IEEE T-IM</i>	Phantom torso + slope	KUKA LWR 4+ 7-DOF	QP-based variable impedance + energy tank for passivity. Z: $K_c \in [100\text{--}500]$ N/m updated per step; XY stiff for scan plane.	Force RMSE 0.50 N scanning (0.88 N static). Matches expert manual scan. Soft-tissue range [125–500] N/m.	Yes — vs manual scan + constant stiffness	10.1109/TIM.2024.3372209
3	Fu et al. 2025 ★ <i>IEEE T-RO</i>	Phantom + user study	KUKA LWR IV+ 7-DOF	QP-VIC + global energy tank + bilateral shared control (Sigma.7 haptic master). $K_{opt} \in [100\text{--}1000]$ N/m. Outer PID force loop.	Force error 0.25 N (best of all). NASA-TLX –54.6% vs position ctrl. Most recent SoA (2025) for hybrid force-impedance.	Yes — vs Fu 2024 + position ctrl	10.1109/TR0.2025.3548493
4	Xie & Yan 2022 <i>Mechanical Sciences</i>	Sim only MATLAB + V-rep	PUMA 560 / KUKA LWR iiwa (sim)	VAHI: adaptive backstepping position (XY) + adaptive impedance variable rate $\sigma=f(e_r,\dot{e}_r)$ in force (Z). $S=\text{diag}(0,0,1,0,0,0)$.	Force RMSE 0.24 N vs 0.91 N classic HI (–74%). Overshoot <1 N. Validated flat/inclined/sinusoidal. Theory architecture: Z force, XY impedance.	Yes — vs HI + AHI	10.5194/ms-13-559-2022
5	Chen et al. 2025 <i>IEEE ICIA</i>	Phantom abdominal	Franka Emika Panda 7-DOF	Cascaded force/position: XY Cartesian impedance + Z (feedforward impedance + PID feedback) + null-space optimisation + dynamic median filter on force.	Force std –24.53% vs pure PID. Mean error flat 0.16 N, curved 0.44 N. Null-space reduces joint fluctuations. Most complete arch (2025).	Yes — vs pure PID (10 trials)	10.1109/ICIA64617

1b Visual Servoing (IBVS) — lesion tracking in the US image

SCOPUS ("robot*") AND ("ultrasound") AND ("visual servoing") AND ("tracking")

#	Authors, Year	Study	Robot	Control method	Key result	Benchmark	DOI
6	Šuligoj et al. 2022 ★ <i>IEEE T-BME</i>	Phantom + bio tissue	Franka Emika Panda 7-DOF	DCF-CSR visual servoing + impedance contact $K_s=\text{diag}[3000,3000,100,300,300,300]$. US vessel segmentation: EPF filter → adaptive threshold → contour. Surface path from centroids + 3D camera point cloud.	Tracking 99.8% at 28–36 FPS; multi-agent 13 FPS with occlusion recovery. Min contact force 2.02 N. Segmentation 100% detection (IOU 0.93). Blueprint for US-surface visual servoing — directly applicable to lesion tracking.	Yes — single/multi-agent, 2 phantom types	10.1109/TBME.2022.3176746

1c Lesion search / Adaptive Trajectory Planning

SCOPUS ("robot*") AND ("breast") AND ("ultrasound") AND ("trajectory" OR "planning")

#	Authors, Year	Study	Robot	Control method	Key result	Benchmark	DOI
7	Tan et al. 2024 <i>IEEE T-ASE</i>	Phantom silicone breast, hyperechoic masses	Custom dual-arm 5-DOF + FUPCD	ACOTG: global offline S-strip scan from 3D point cloud + online local tracking (3rd-order FIR motion decomposition). Detection YOLOv4 + KCF. Closed-loop admittance force ctrl.	Horizontal error <±1 mm , depth ±10%, force 5 N ±15%. ACOTG <1 μs/step. Global trajectory <4 s. Detection 99.05%. Reference: global search + local tracking.	Yes — direct vs ACOTG, update intervals	10.1109/TASE.2023.3262844

Task 1 takeaway for the thesis control loop. The supervisor brief (23/02) specifies Cartesian Impedance Control compliant along the surface normal + minimal IBVS on the lesion centroid. The literature converges on this: **Ipsen 2021** gives the canonical asymmetric-stiffness recipe ($k_z \rightarrow 0$, $k_{x,y}$ stiff) with a breast force window of 3–5 N; **Šuligoj 2022** is the directly transferable US visual-servoing blueprint; **Tan 2024** provides the global-scan-then-local-track architecture. Breast soft-tissue stiffness consistently falls in [125–500] N/m.

02 Ultrasound-Guided Breast Biopsy

CONSTRAINED · SAFETY-CRITICAL · QUASI-STATIC

2a Complete robotic systems (architecture + control)

SCOPUS ("robot*") AND ("breast") AND ("biopsy") AND ("image-guided" OR "ultrasound-guided")

#	Authors, Year	Study	Robot	Control method	Key result	Benchmark	DOI
8	Ferrari et al. 2023 ★★ <i>IEEE RA-L</i>	Phantom CIRS Model 073, 15 lesions MRI+US	UR5e 6-DOF	5-phase autonomous: (1) MRI segmentation → centroid (2) ICP registration RGB-D 360° (3) surface approach (4) preloading 2 N parallel force-position (5) Neo-Hookean FEM compensation (6929 tetrahedra, real-time, Signorini contact).	Median lesion MSE 3.3 mm at 2 N (vs radius 3.6 mm). FEM gives ~48% error. Robotic 3.34 mm vs manual 4.2 mm. Optimal preload 2 N. Most advanced complete SoA — same phantom as the thesis. Code on GitLab.	Yes — ±DC, vs freehand, 3 preload levels	10.1109/LRA.2023.3237499
9	Liang et al. 2010 <i>Ultrasound Med. Biol.</i>	Phantom turkey tissue, calcf. + cyst	Gantry III Cartesian 3-DOF	Open-loop position + 3D US thresholding. Hyperechoic → above-threshold voxels → centroid; anechoic → below + 1 mm proximity filter. Volume → MATLAB segmentation → coords → motion (no feedback).	Needle-touch rms 1.15 mm (1.45% path). Success 5/5 calcf + 5/5 cyst. Time ~1 min / ~3 min. First fully autonomous US → robot biopsy — thresholding sufficient for reliable 3D coords.	Yes — vs commercial stereotactic (~2 mm)	10.1016/j.ultrasmedbio.2009.08.014
10	Mallapragada et al. 2011 <i>IEEE/ASME T-Mech</i>	Phantom deformable, heterogeneous	Custom 3 fingers + US + guide, 6 DOA	Target manipulation: 3 fingers move tumour toward needle path. PI ctrl (P outer + PI inner velocity) on US centroid. Sinusoidal plane search if lost. Segmentation: region → hist eq → threshold → median → blob centroid.	Targeting RMSE 0.17 mm (max 0.40) over 13 trials. Positioning 37 s. Error ~1 order below needle Ø (2.5 mm). First system with autonomous US centroid tracking as control feature.	Yes (implicit) — vs freehand/stereotactic	10.1109/TMECH.2010.2068575
11	Welleweerd et al. 2020 <i>IJCARS</i>	Phantom wax cuboid + EM tracker	KUKA MED 7 R800 7-DOF, medical	3-DOF motorised needle guide (IK toward target in US plane) + pneumatic depth stop (laser, 20 µm, valve lock) + stereo camera + US holder. Pre-op ICP on stereo markers. Probe at 45°.	In-air guide accuracy y 0.1±0.36, z 0.3±1.5 mm. Depth stop 0.100 mm . Lock force 3.5–6 N. Full system Euclidean 3.21 mm. EE design reference (forces, workspace, 45° angulation). MURAB project.	Yes — vs literature 0.25–3.44 mm	10.1007/s11548-020-02122-1

2b Force/position control during needle insertion

#	Authors, Year	Study	Robot	Control method	Key result	Benchmark	DOI
12	Shi et al. 2002 <i>IEEE T-CST</i>	Sim only MATLAB	3R planar + linear actuator (PUMA 200 red.)	Stereo-fluoroscopic 3D path + adaptive trajectory ctrl. PD + friction (B) + stiffness (K) compensation; adaptive online estimation of unknown B,K (Barbalat → asymptotic). Needle bending avoidance via velocity/accel limits.	Asymptotic tracking proved in 4 scenarios. Lesion displacement → online path update eliminates error in 0.5 s. Imaging-delay constraint derived. First formalisation of tissue impedance compensation for biopsy.	No — analytical convergence only	10.1109/87.998010

2c Tissue deformation management (safety-critical)

#	Authors, Year	Study	Robot	Method	Key result	Benchmark	DOI
13	Tagliabue et al. 2019 <i>IJCARS</i>	Phantom CIRS, 10 lesions + gel	No robot (computational + freehand)	Position-Based Dynamics (PBD) real-time deformation. Offline genetic-algorithm calibration + fine-tuning on target phantom. Compared vs Neo-Hookean FEM (SOFA) and rigid model.	Mean error <7.26 mm (deform 10–30 mm). Speed 6.99 s vs 16.37 s FEM (2.3× faster). Fine-tuning ~24%. ≥50% more accurate than rigid. Real-time alternative to FEM.	Yes — vs FEM (SOFA) + rigid	10.1007/s11548-019-01997-z
14	Groenhuis et al. 2018 <i>IJCARS</i>	Phantom 4 PVC, MRI prone/supine	No robot (MURAB project)	Analytical Young's modulus from gravity-induced deformation: D = β·g/E from paired prone/supine MRI. No FEM mesh — layer-by-layer analytical integration. Compared vs SOFA, FEBio.	Deviation vs FEM ≤9% (cylindrical), ≤18% (breast). Compute <1 s vs minutes. E = 4364–9677 Pa (biologically plausible). 100–1000× faster. Provides E input for FEM/PBD models.	Yes — vs SOFA + FEBio, 5 resolutions	10.1007/s11548-018-1803-x

2d General review — read first

#	Authors, Year	Study	Scope	Content	Key findings	Benchmark	DOI
15	Kulkarni et al. 2019 ★ <i>Ann. Biomed. Eng.</i>	Review ~80 systems 2008–19	MRI / CT / US / multi-imager	Taxonomy of all actuator types (piezo, pneumatic, DC...) and control schemes (PID, SMC, impedance, co-manipulation). Compatibility tests per modality.	FDA approves only manual needle insertion (robot = positioning). US = least constrained modality, DC motors OK. Trends: body-attached, multi-imager, force+image feedback. Essential taxonomy for the SoA chapter.	N/A — systematic review	10.1007/s10439-019-02319-9

Task 2 takeaway for the thesis. **Ferrari 2023** is the anchor reference — same CIRS Model 073 phantom, full autonomous pipeline (segmentation → registration → approach → preloading → FEM deformation compensation), median error 3.3 mm below lesion radius, code public. The historical line (**Liang 2010**, **Mallapragada 2011**) proves that simple US thresholding + centroid IBVS is sufficient for reliable 3D targeting — matching the thesis localisation goal. Deformation is bracketed by FEM (Ferrari) and faster PBD (**Tagliabue 2019**), with tissue elasticity E from **Groenhuis 2018**. Regulatory frame (**Kulkarni 2019**): robot positions, clinician inserts.

15 papers — all read in full · Scopus + IEEE Xplore · Curated for thesis Moro · Robotic CT-guided breast biopsy · Extraction per Lamon brief 17/12/2025